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SUPERSEDING
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16 JUNE 1944

MILITARY SPECIFICATION

TANK, FUEL, FOR PORTABLE FLAME THROWER, M2A1

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of fuel tank.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-P-106 —Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.

MIL-L-3150 —Lubricating-Oil, Preservative, Medium.

MIL-P-10577 —Packaging of Components and Subassemblies, Domestic Shipment to Assembly Plants.

MIL-F-11385 —Flame Thrower, Portable, M2A1.

U. S. ARMY

57-206 —Welding of Constructional Steel, Metal-Arc, Manual or Machine; General Specifications For.

57-206-21 —Welding of Constructional Steel; Resistance, Spot.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-10 —Surface Roughness, Waviness, and Lay.

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking of Shipments.

DRAWINGS

CHEMICAL CORPS

C81-1-481 —Tank, Fuel, Portable Flame Thrower, M2A1—Bill of Material—Sheet 1 of 2 Sheets.

C81-1-529 —Tank, Fuel, Portable Flame Thrower, M2A1—Bill of Material—Sheet 2 of 2 Sheets.

B81-1-879 —Tank and Valve, Pressure—Assembly and Bill of Material.

B81-1-883 —Shaft, Valve—Assembly and Bill of Material.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.**—All materials and components used in the manufacture and assembly of the tank shall conform to the specifications and drawings forming a part of this specification.

3.2 **Manufacture and assembly.**—The tank shall be manufactured and assembled as

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specified on Drawing D81–1–432 and subsidiary drawings listed on Drawings C81–1–481, and C81–1–529. Surface roughness, waviness, and lay specified by symbols on detail drawings shall comply with the applicable classes specified in Standard MIL–STD–10.

3.3 Components.

3.3.1 *Pressure tank and valve.*—The pressure tank and valve shall be assembled as specified on Drawing B81–1–879. The entire assembly shall not leak when tested as specified in 4.4.1 and 4.4.1.1.

3.3.1.1 *Flexible valve shaft.*—The flexible valve shaft, Drawing E81–1–470 which is listed on Drawing B81–1–883, shall not fail when tested as specified in 4.4.1.2.

3.3.2 *Fuel tank.*

3.3.2.1 *Leakage.*—The assembly, Drawing D81–1–483, which is listed on Drawing C81–1–481, shall not leak when tested as specified in 4.4.2.1.

3.3.2.2 *Durability.*—The assembly shall not burst when tested as specified in 4.4.2.2.

3.3.3 *Safety head.*

3.3.3.1 *Leakage.*—The assembly, Drawing E81–1–561 which is listed on Drawing C81–1–481, shall not leak when tested as specified in 4.4.5.1.

3.3.3.2 *Bursting.* — The diaphragm, when assembled in the safety head, shall burst when tested as specified in 4.4.5.2.

3.4 Assembly leakage. — The following joints shall not leak when tested as specified in 4.4.3.

3.4.1 *At 2100 p.s.i.*

(a) Between the regulator tube, Drawing E81–1–1794, and the socket, Drawing A81–1–884, both drawings listed on Drawing C81–1–481.

(b) Between the regulator tube and the pressure regulator, Drawing B81–1–778 which is listed on Drawing C81–1–481.

(c) Between the socket and the plug, Drawing E81–1–885 which is listed on Drawing B81–1–879.

3.4.2 *At 375 p.s.i.*

(a) Between the safety valve, Drawing B81–1–1796 which is listed on Drawing C81–1–529, and the regulator.

(b) Between the top of the safety valve and the diffusion pipe, Drawing B81–1–494 which is listed on Drawing C81–1–481.

(c) Between the bottom of the safety valve and the safety head.

(d) Between the tank coupling, Drawing D81–1–4581 which is listed on Drawing C81–1–529, and the coupling plug, Drawing E81–1–514 which is listed on Drawing C81–1–481.

(e) Between the tank coupling and the pipe connector, Drawing D81–1–483 which is listed on Drawing C81–1–481.

(f) Between the filling plug, Drawing C81–1–2450 which is listed on Drawing C81–1–529, and the plug seat, Drawing A81–1–488 which is listed on Drawing C81–1–481.

3.5 Welding.

3.5.1 *Metal-arc.* — All metal-arc welding shall conform to Specification 57–206.

3.5.2 *Spot resistance.*—All spot resistance welding shall conform to Specification 57–206–21.

3.6 Performance.

3.6.1 *Tanks shipped to flame thrower assembly plants.*—The fuel tank, when assembled into a complete flame thrower, shall

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meet the performance requirements when tested as specified in 4.4.4.1.

3.6.2 *Tanks shipped as spare parts.*—The fuel tanks to be shipped as spare parts shall show an air-flow resistance of not more than 8 p.s.i. when tested as specified in 4.4.4.2.

3.7 **Workmanship.** — All work shall be done in a neat and workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.

4.1.1 *Assembly.*—A lot shall consist of the assembled tanks produced by one manufacturer, from the same materials, and under essentially the same manufacturing conditions. The inspection lot for nondestructive inspection, within the above limitations, shall be designated by the inspector.

4.1.2 *Safety head.*—A lot shall consist of the completely assembled safety heads with the diaphragms died out of one sheet of rolled metal measuring approximately one square foot in area. The inspection lot for nondestructive inspection, within the above limitations, shall be designated by the inspector.

4.2 Sampling.

4.2.1 *For nondestructive inspection.*—Sampling for nondestructive inspection shall be conducted in accordance with the authorized classification of defects (4.3.3) and Standard MIL–STD–105.

4.2.2 *For durability tests.*—One representative fuel tank assembly, Drawing D81–1–483 which is listed on Drawing C81–1–481, shall be taken from each lot of 200 or less for testing as specified in 4.4.2.2 (see 6.3).

4.3 Inspection.

4.3.1 *Components.* — The inspector shall ascertain that all materials, components, or subassemblies of the tank covered by separate drawings, specifications, or standards are in accordance with their applicable drawings, specifications, or standards prior to final acceptance of the tank.

4.3.2 *Packaging, packing, and marking.*— The inspector shall inspect the packaging, packing, and marking of the tank for compliance with section 5.

4.3.3 *Classification of defects.* — Inspection of the tank shall be conducted as follows:

4.3.3.1 *Tank, fuel, portable flame thrower, M2A1—assembly.* — (Dwg. D81–1–482 is listed on Drawing C81–1–481)

Categories and defects	*Gage or other method of inspection*
Critical	
A. Pressure tank and valve, pressure	4.4.1 and 4.4.1.1
B. Fuel tank pressure	4.4.2
C. Assembly pressure	4.4.3.1, 4.4.3.2, and 4.4.3.3.
D. Performance	4.4.4.1 and 4.4.4.2.
Major A AQL 0.65 percent	
*1a. Safety head leakage	4.4.5.1
Major AQL 1.0 percent	
1. Component missing or incorrectly assembled.	Visual
2. Component damaged	Visual
3. Marking incorrect	Visual
*4. Safety head thread (⅝"-18NF-2) major and pitch diameter	E24–1–194 and B24–1–99
Minor AQL 2.5 percent	
*51. Safety-head dimensions incorrect	Comparison
52. Threads not luted	Visual
53. Marking missing or illegible	Visual
54. Coating missing, inadequate, or incorrect where required.	Visual
Incidental AQL 10 percent	
101. Contamination	

* Safety head inspection shall be made on the basis of prime Chemical Corps procurement for test and dimensional requirements on Drawing E81–1–561 which is listed on Drawing C81–1–481 and this specification. Following the diaphragm bursting test the diaphragm shall be replaced by a new diaphragm.

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4.4 Tanks.

4.4.1 *Pressure tank and valve.*—The entire assembly shall be tested with a relatively oil-and-water-free internal gas pressure of not less than 2,100 p.s.i. and show no bubbles when immersed in water. Test for leakage for 5 minutes with the valve closed, and then for an additional 5 minutes with the valve open and the plug cap removed in both cases.

4.4.1.1 *Pressure tank valve.*—The pressure tank valve, Drawing B81–1–439 which is listed on Drawing B81–1–879, shall be subjected to a relatively oil-and-water-free internal gas pressure of not less than 2,100 p.s.i. applied through the inlet and show no bubbles when immersed in water. Test for leakage for 5 minutes with the valve open and with the valve closed (see 6.4).

4.4.1.2 *Flexible valve shaft.*—The flexible valve shaft shall withstand a 100 pound-inch torque in both directions. The torsional deflection in a winding direction shall not exceed 6° per foot of shaft and in an unwinding direction shall not exceed 7° per foot of shaft both at 30 pound-inch loads.

4.4.2 *Fuel tank.*

4.4.2.1 *Leakage.*—Each complete fuel tank assembly shall be subjected to an internal hydrostatic pressure of 625 p.s.i. without leaking or deforming.

4.4.2.2 *Durability.*—The assembly shall be subjected to an internal hydrostatic pressure of 900 p.s.i. The lot represented shall be rejected if any specimen bursts at less than 900 p.s.i. Any deformed tank after test shall not be reused.

4.4.3 *Assembly leakage.*

4.4.3.1 *2100 p.s.i. internal gas pressure.*—The following joints shall be tested with a relatively oil-and-water-free internal gas pressure of not less than 2100 p.s.i. by immersing in water for a period of 5 minutes and observing for leaks:

(a) Between the regulator tube and the socket.
(b) Between the regulator tube and the regulator.
(c) Between the socket and the plug.

4.4.3.2 *375 p.s.i. internal gas pressure.*—The following joints shall be tested with a relatively oil-and-water-free internal gas pressure of not less than 375 p.s.i. by immersing in water for a period of 5 minutes and observing for leaks.

(a) Between the safety valve and the regulator.
(b) Between the top of the safety valve and the diffusion pipe.
(c) Between the bottom of the safety valve and the safety head.
(d) Between the filling plug and the plug seat.

4.4.3.3 *375 p.s.i. internal hydrostatic pressure.*—The following joints shall show no leaks when tested under a 375 p.s.i. internal hydrostatic pressure:

(a) Between the tank coupling and the coupling plug.
(b) Between the tank coupling and the pipe connector.

4.4.4 *Performance.*

4.4.4.1 *Tanks shipped to flame thrower assembly plants.*—The fuel tank, when assembled into a complete flame thrower, shall be tested for performance as specified in Specification MIL–F–11385.

4.4.4.2 *Tanks shipped as spare parts.*—Tanks shipped as spare parts shall be tested as follows: Connect the diffuse line to a source of compressed air. Open the pressure tank valve and remove the fuel filling plugs. Adjust the air-flow rate to 14 ± 0.5 cubic feet per minute and observe the resistance (see 6.5).

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4.4.5 *Safety head.*

4.4.5.1 *Leakage.*—The assembly shall be tested as follows: Attach the male-thread end of the safety head to a source of adjustable air pressure. Submerge the safety head in water and gradually apply an air pressure until 400 p.s.i. is obtained. Hold for a period of 1 minute and observe for leaks.

4.4.5.2 *Bursting.* — Three specimen diaphragms when assembled in safety heads taken from the sample passing the leakage test shall be tested for bursting as follows: Attach the male-thread end of each head to a source of adjustable air pressure. Submerge the heads in water or place behind a suitable barricade and gradually apply the air pressure until the diaphragm bursts. If any diaphragm does not burst at a pressure between 525 and 625 p.s.i., the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 **Packaging and packing.**

5.1.1 *For shipment to assembly plants.*—The tank assembly shall be prepared for shipment to flame-thrower assembly plants in accordance with Specification MIL–P–10577.

5.1.2 *For shipment as spare parts.*—The tank assembly shall be prepared and packed for shipment as spare parts as follows: Flush fuel tanks *only* with medium preservative lubricating oil conforming to Specification MIL–L–3150. Shut all valves and hermetically seal all openings. The fuel tank shall be packed in a style 2, 2½, or 4 nailed wood box conforming to Specification JAN–P–106 and be braced and blocked in the box to prevent shifting and damage. The box shall have the following approximate inside dimensions: 26½ inches by 20⅜ inches by 11⅝ inches. The top, bottom, and sides shall be made of ⅞-inch lumber, the ends of ⅝-inch lumber, and the cleats of ⅝- by 1¾-inch lumber. The box shall be strapped in accordance with appendix I to Specification JAN–P–106.

5.2 **Marking.**—In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.**—The item covered by this specification is intended for use as a component of the M2A1 portable flame thrower.

6.2 **Ordering data.** — Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether shipment to assembly plants or as spare parts is required.

6.3 **Sampling and testing.**—When the contractor, consistently produces high quality material and operates under a system of quality control acceptable to the Government, the Government at its discretion, may modify in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

6.4 **Pressure-tank valve.**—At the option of the inspector the leakage test may be made either on the valve or when testing the pressure-tank and valve assembly.

6.5 **Air-flow resistance of fuel tanks.**—The following arrangement has been found satisfactory for testing the air-flow resistance of the fuel tanks: A source of compressed air is connected to a pressure regulator set to deliver air at 350 p.s.i. The air is metered

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through a 0.046-inch thick, 0.053 ± 0.001-inch diameter orifice in a ¼-inch pipe coupling and the air passed through sufficient pipe or tubing to produce a 5 p.s.i. back pressure on a pressure gage on the downstream side of the orifice and at the end of the pipe or tubing. The end of the downstream piping is then connected to the diffuse pipe of the flame-thrower tank and the air resistance of the fuel tank is read as the increase in back pressure on the gage on the downstream side of the orifice.

6.6 Caution.—Due to the hazards inherent in testing with high pressure air or gases, manufacturers are hereby notified that every precaution possible should be taken for the safety and protection of the personnel engaged in testing the tank.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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